Trade-offs for Performance in Reporting:

> What techniques can we use to make reports run faster, and what do we give up in return?

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1. Indexes

Benefit

Reports become faster.

```sql
CREATE INDEX idx_order_date
ON orders(order_date);
```

Trade-off

Faster SELECT
Slower INSERT/UPDATE/DELETE
More disk space

text

```
More Indexes
    ↓
Faster Reports
    ↓
Slower Writes
```

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2. Materialized Views

Benefit

Store precomputed report data.

```sql
CREATE MATERIALIZED VIEW sales_summary AS
SELECT region,
       SUM(amount)
FROM sales
GROUP BY region;
```

Trade-off

Very fast reporting
Data may become stale

```text
Fast Report
    ↓
Old Data Until REFRESH
```

```sql
REFRESH MATERIALIZED VIEW sales_summary;
```

---

3. Denormalization

Instead of joining many tables:

Normalized

```text
Orders
Customers
Products
```

Report requires multiple joins.

Denormalized

```text
Orders
 ├─ customer_name
 ├─ product_name
```

Trade-off

Faster reports
Data duplication
More storage
Update anomalies

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4. Replicas (Read Replicas)

Benefit

Run reports on replica.

```text
Primary DB
      |
      V
Replica DB
```

Trade-off

- Reporting doesn't affect production
- Replica lag may occur

```text
Fast Reporting
       vs
Slightly Old Data
```

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5. Partitioning

Split large tables.

text
sales_2024
sales_2025
sales_2026

Trade-off

Faster queries on specific partitions
More complex administration

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6. Aggregation Tables

Store summarized data.

sql
daily_sales

instead of calculating from millions of rows every time.

Trade-off

Fast reports
Extra ETL/maintenance

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7. Caching

Store report results.

```text
User Request
      ↓
Cache
      ↓
Database
```

Trade-off

Very fast
Cache can become outdated

Technique	Gain	Trade-off
Indexes	Faster queries	Slower writes
Materialized Views	Faster reports	Stale data
Denormalization	Fewer joins	Data redundancy
Replication	Offload reporting	Replica lag
Partitioning	Faster scans	More complexity
Aggregation Tables	Quick summaries	Maintenance overhead
Caching	Instant response	Data freshness issues